COLLEGE OF ENGINEERING
GRADO DEPARTMENT OF INDUSTRIAL AND SYSTEMS ENGINEERING
VIRGINIA TECH

GRADUATE PROGRAMS · ONLINE OPEN HOUSE

Faculty and Research

Meet the people behind the work



November 12, 2026

Five program areas · 45 faculty profiles

Explore research questions, recent publications and faculty recognition. Follow the links to learn more and start a conversation.

Faculty photograph: Virginia Tech ISE, Fall 2026

EXPLORE THE DEPARTMENT

Find a research connection

Start with a question you want to investigate. Then meet the faculty whose methods and interests connect to it.

Human Factors Engineering and Ergonomics 03

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Operations Research 20

AI, Autonomy, Machine Learning, and Robotics 26

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Use this booklet

Each profile introduces a faculty member's work, highlights a publication, recognition or research focus, and suggests a question for a conversation. Click the links for source material and further reading.

Follow the connections

Many faculty work across areas. Each person has one full profile; area introductions and the index point to additional connections. Teaching and practice profiles describe those roles separately.

Publication and award dates are stated where available. Preprints and modeled results are labeled. Profiles describe research interests and roles; contact faculty directly about current advising and funding opportunities.

Prepared for the November 2026 open house. Sources include department program pages, faculty and laboratory websites, publication records and university announcements, checked November 11, 2026. Portraits are from official faculty profiles. Full faculty directory · Graduate manual

PROGRAM AREA · 01

Human Factors Engineering and Ergonomics

Design systems around the people who use them.

Human Factors Engineering and Ergonomics studies the fit between people and the technologies, tasks and environments they encounter. Graduate study connects experimental methods with the measurement of human capabilities and limitations. Research can begin with a physical challenge, such as lifting or recovering balance, or a cognitive challenge, such as interpreting a display and responding to an unexpected event.

The area spans sensory, physical, cognitive and organizational dimensions of work. Students may investigate hearing and communication, biomechanics and fatigue, human information processing, or the way teams and organizations coordinate their activities. This breadth makes it possible to connect laboratory evidence with the design and evaluation of real systems.

At Virginia Tech, the faculty community includes work on immersive interfaces, auditory systems, driver behavior, workplace safety, assistive technologies and human-robot interaction. The profiles that follow show different ways to approach a common question: what should change in the design when we understand the user more fully?

Methods you could use

Human-subject experiments; usability evaluation; biomechanical measurement; wearable sensing; cognitive work analysis; statistical modeling.

Questions to explore

A possible research question is how an assistive device changes both physical effort and task performance. Another is how a warning or augmented display affects attention in a demanding environment. The right measures depend on the people, task and setting.

Graduate preparation and pathways

The department describes doctoral concentrations in Cognitive Engineering, Human Factors Engineering and Product Design, and Occupational Ergonomics and Safety. Explore the graduate manual with an advisor to connect your interests to the appropriate degree plan.

Faculty connections
Dickerson, Kleiner and Lau also connect to Management and Systems Engineering. Jeon and Lim are listed with A2MR. These connections are cross-referenced in the faculty index.

Program overview and affiliated laboratories · Current graduate manual



Deborah E. Dickerson

Associate Professor · CIH, CSP

Preventing harm through product and system design

Dickerson directs the Product and System Safety Laboratory and works across human factors and management systems. Her research asks how design can prevent hazards before people encounter them. Topics include warnings and instructions, failure-mode analysis, risk perception, trust in technology and infrastructure safety. Her background in industrial hygiene and safety connects engineering design with occupational health. Research can involve evaluating how people interpret information as well as how organizations identify and manage risk.

RESEARCH FOCUS

The laboratory's projects have examined chemical-warning comprehension, product instructions, acceptance of innovations and risk-based management of public infrastructure. These questions connect user experience with the design of safer products and work systems.

A conversation starter
How can we redesign a product or process so that safety does not depend only on a warning?

Faculty profile · Lab / personal site · Publications



Joseph L. Gabbard

Professor · Hal G. Prillaman Professor

Augmented reality designed around human perception

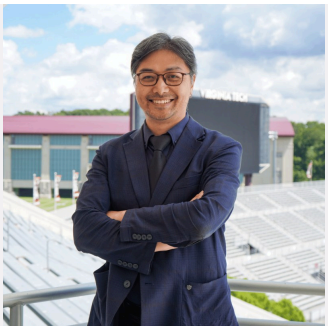
Gabbard studies augmented and virtual reality, human-computer interaction and usability engineering. His work considers how people perceive, understand and act on digital information embedded in their surroundings. This connects interface design with experiments that evaluate user performance. For prospective students, the research offers a way to combine interactive technology with questions about attention, perception and useful support for real tasks.

RESEARCH AND RECOGNITION · 2026

Virginia Tech named Gabbard to the Hal G. Prillaman Professorship. A September 2026 announcement also lists him with Doug Bowman on an interdisciplinary project to enable low-friction intelligent extended reality.

A conversation starter
When does an augmented display help someone perform a task, and when does it add effort?

Faculty profile · Publications · Highlight source



Myounghoon “Philart” Jeon

Professor

Human interaction through sound, emotion and robotics

Jeon’s research connects human-computer and human-robot interaction with sound, music, emotion and assistive technology. His projects examine how an interaction feels and how it affects behavior, accessibility and performance. Applications include vehicle interfaces, immersive experiences and collaborative technologies. This work brings experimental psychology together with design and computing, offering students opportunities to study interactions that extend beyond a conventional screen.

RECENT WORK AND RECOGNITION

His 2024 publications include studies of audio augmented reality and in-vehicle intelligent agents. The College of Engineering’s 2026 Dean’s Awards list recognizes Jeon for Excellence in Research.

A conversation starter
How can sound or emotional feedback make an intelligent system easier to understand and use?

Faculty profile · Lab / personal site · Publications · Highlight source



Sunwook Kim

Research Associate Professor

Measuring physical demands and ergonomic risk

Kim studies occupational biomechanics, exposure assessment and the control of posture and balance. His research connects the measurement of human movement with questions about safer work and more inclusive technology. The program-area description also identifies ergonomic interventions and inclusive workplace technology among his interests. Students drawn to physical ergonomics can explore how to measure demands, evaluate interventions and account for differences among users.

INCLUSIVE ROBOTICS PROJECT · 2025

Kim leads an NSF-supported team investigating collaborative robots that could provide personalized support to autistic workers. The project connects human-centered manufacturing with feedback, autonomy and well-being, involving collaborators including Myounghoon Jeon and Sol Lim.

A conversation starter
Which measurements best show whether a workplace intervention reduces physical demands?

Faculty profile · Publications · Highlight source



Charlie Klauer

Associate Professor

Understanding driving risk in everyday conditions

Klauer studies driver distraction, fatigue and novice teen drivers. Her work connects human factors with transportation safety, using evidence about driver behavior to understand when risk increases. Her faculty profile describes leadership in teen risk and injury prevention at the Virginia Tech Transportation Institute. Students interested in transportation can connect behavioral measurement, statistical analysis and the design of interventions for safer driving.

SELECTED RESEARCH

Her publications examine distraction and crash risk, differences between novice and experienced drivers, and the contribution of naturalistic driving research to prevention. Together, these studies connect driver behavior with the conditions that contribute to crashes.

A conversation starter
How can observations of real driving help identify interventions that matter outside a simulator?

Faculty profile · Publications



Brian M. Kleiner

Vorster-Kleiner Director of the Myers-Lawson School of Construction · Professor

Safety across people, technology and organizations

Kleiner studies work systems at the organizational scale, including occupational safety, macroergonomics and the allocation of functions between people and automation. He directs the Myers-Lawson School of Construction and connects human factors with construction safety and well-being. His work considers how technology, management practices and collaboration shape the conditions in which people work. It crosses the human factors and management-and-systems communities.

RECOGNITION · 2025

His faculty profile records induction into the National Academy of Construction in 2025. His earlier work includes studies of construction safety climate and maintaining patient care during electronic-health-record outages.

A conversation starter
How should organizational design change when a new technology alters responsibility for safety?

Faculty profile · Publications · Highlight source



Nathan Lau

Associate Professor

Interfaces for complex and safety-critical work

Lau studies cognitive engineering, situation awareness, interface design and human performance. His work examines how operators understand complex systems and how displays can support effective action. Applications described by his faculty profile include nuclear process control, remotely piloted aircraft operations and medical technology. He connects human-subject evaluation with systems analysis, bridging human factors and management and systems engineering.

RESEARCH FOCUS

Lau's work on ecological interface design and cognitive work analysis addresses the relationship between a system's underlying constraints and the information presented to its users. His research also examines how situation awareness can be assessed reliably.

A conversation starter
What must an operator understand about a system before a display can help them make a safe decision?

Faculty profile · Lab / personal site · Publications



Kichol Lee

Research Assistant Professor

Hearing and communication in demanding environments

Lee's research focuses on auditory situation awareness, hearing-related systems, communication and military ergonomics. He also studies product design and evaluation. These interests connect the properties of an acoustic environment or device with a person's ability to notice, interpret and respond to important information. Students interested in perception can explore the engineering of systems that must support communication while functioning in difficult real-world conditions.

RESEARCH APPLICATION · 2024

His work on the Portable Auditory Situation Awareness Test (PASAT) evaluates auditory situation awareness with and without hearing protection. It connects hearing protection and equipment evaluation with the ability to detect and interpret sounds in operational environments.

A conversation starter
How can a listening or communication system preserve the cues that people need to understand their surroundings?

Faculty profile · Highlight source



Sol Lim

Associate Professor · Chachra Faculty Fellow

Wearable sensing for accessible and safer activity

Lim combines ergonomics, biomechanics and predictive modeling with wearable technology. Her interests include occupational health, healthcare ergonomics, disability and inclusive design. This research connects measurements of movement and physical demand with the design of tools, environments and training approaches. The program-area page also highlights haptics, accessibility and remote physical training, linking physical ergonomics to emerging human-technology interactions.

RECOGNITION · 2026

The Board of Visitors' April 2026 appointment materials name Lim to the Vinod and Ranjana Chachra Fellowship in Industrial and Systems Engineering. Her research includes wearable sensing and the needs of people with different mobility capabilities.

A conversation starter
How can wearable measurements guide a design that works for people with different physical abilities?

Faculty profile · Lab / personal site · Publications · Highlight source



Michael L. Madigan

Professor · Senior Grado Faculty Fellow

Biomechanics of balance and fall prevention

Madigan investigates slips, trips and falls, occupational biomechanics and physical performance. His work considers how aging, fatigue and other individual characteristics influence movement and injury risk. The research combines biomechanical measurement with questions that matter for daily life and work. Students can study how a loss of balance occurs, how people recover and how training or design changes might reduce the risk of falling.

RESEARCH SUPPORT · SEPTEMBER 2026

Virginia Tech's grants listing identifies Madigan as principal investigator on an NIH National Institute on Aging project evaluating perturbation balance training for older adults with impaired balance, including laboratory and real-world trip-related outcomes.

A conversation starter
How can a laboratory balance-training method be evaluated for its effect on everyday falls?

Faculty profile · Lab / personal site · Publications · Highlight source



Maury A. Nussbaum

John Grado Professor · Graduate Program Director

Physical ergonomics and injury prevention

Nussbaum studies the physical demands of work, injury prevention, muscle fatigue and balance. His research combines biomechanics and work physiology with the evaluation of equipment, tasks and ergonomic interventions. It considers both the demands of an activity and differences in people's capabilities. The HFEE area also highlights exoskeleton research among his interests. He serves as the department's Graduate Program Director and Assistant Department Head.

RECOGNITION · 2026

Nussbaum received the David F. Baker Distinguished Research Award from the Institute of Industrial and Systems Engineers in 2026, recognizing substantial research contributions to industrial engineering.

A conversation starter
How can we evaluate an assistive device for both reduced physical demand and unintended effects?

Faculty profile · Lab / personal site · Publications



Rafael Patrick

Assistant Professor

Perception and the design of human-machine interfaces

Patrick studies how people sense, interpret and use information when interacting with technology. His interests include applied psychophysics, auditory situation awareness, human information processing and user-centered design. The work links fundamental questions about perception to product evaluation and multimodal user experience. Students interested in experiments can examine how an interface's sensory cues affect comprehension, experience and performance.

RESEARCH COLLABORATION · SEPTEMBER 2026

Patrick contributed audio expertise to a Virginia Tech–Zoox study investigating how vehicle sounds communicate with pedestrians. He helped ensure controlled playback and accurate audio capture, connecting perceptual measurement with the design of automated-vehicle interactions.

A conversation starter
How should we measure whether a sensory cue is perceptible, meaningful and useful to the user?

Faculty profile · Publications · Highlight source

PROGRAM AREA · 02

Manufacturing Systems Engineering

Connect physical processes, data and production decisions.

Manufacturing Systems Engineering considers how products are made and how the surrounding production system should operate. Graduate work connects manufacturing technologies with system integration, economic analysis and operations. A project may focus on the physics of a process, the interpretation of sensor data or the movement of products and materials through a facility.

The department's research includes biomanufacturing, sensing and biosensing, quality assurance, process monitoring and control, automation, and the security of connected manufacturing systems. These interests reach across sectors including aerospace, transportation, biomedical devices, pharmaceuticals and semiconductors.

The faculty profiles show several routes into this work. Some groups combine experiments with machine learning and feedback control. Others develop production and logistics models or study sustainable systems that incorporate reuse and remanufacturing. The area also connects with university-wide manufacturing collaborations and the department's industry-facing Learning Factory.

Methods you could use

Process experiments; statistical modeling; sensing and signal analysis; optimization; simulation; control; computer vision; digital twins.

Questions to explore

A possible project could connect process measurements to early defect detection, or compare how a production plan changes when returned products share equipment with new products. These examples link a local technical decision to system-level performance.

Graduate preparation and pathways

The program identifies simulation and optimization as important preparation. Students without that background should discuss suitable coursework with an advisor. The current graduate manual gives the requirements for each degree pathway.

Faculty connections
Sarin's profile appears in Operations Research; L'Afflitto's appears with A2MR. Cherbaka, Coleman, Nguyen and Savage appear in the teaching and professional practice section.

Program overview and affiliated laboratories · Current graduate manual



Kimberly P. Ellis

Associate Professor

Production and logistics systems that work in practice

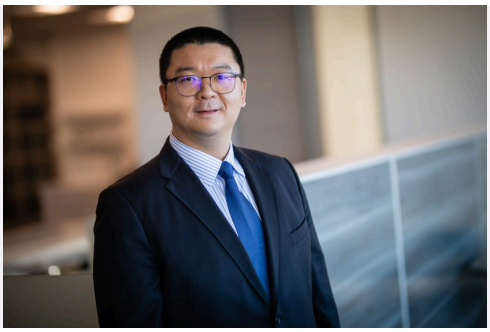
Ellis applies operations research to the design and analysis of manufacturing and logistics systems. Her research includes production planning, process planning and the coordination of resources across a system. She connects mathematical models with practical operating decisions and directs Virginia Tech's site of the Center for Excellence in Logistics and Distribution. Her recent publications also connect quantitative systems analysis to drinking-water access and affordability.

RECENT PUBLICATION · 2026

A paper with collaborators in the International Journal of Production Research analyzes inspection allocation in genetic manufacturing using nonlinear programming. It illustrates how optimization can guide where quality checks are placed within a production process.

A conversation starter
Where should a production system inspect quality, and how does that choice affect the whole process?

Faculty profile



Ran Jin

Professor

Data science for connected manufacturing

Jin studies how engineering knowledge and data analytics can improve manufacturing processes and networks. His research combines statistical modeling, sensing, optimization and control. The Data Science and Visualization Lab also works on industrial internet systems, computing at the edge and human-machine collaboration. Students can connect mathematical and statistical methods to questions about data quality, process monitoring and decisions made across connected manufacturing systems.

CURRENT PROJECT · 2025–26

His laboratory lists a NIST project on AI-guided domain adaptation and generalization for industrial robots. The project addresses the challenge of using learned methods beyond the particular conditions represented in their original data.

A conversation starter
How can a manufacturing model remain useful when equipment, operating conditions or data sources change?

Faculty profile · Lab / personal site · Publications · Highlight source



Blake N. Johnson

Professor

Biomanufacturing, biosensing and autonomous experiments

Johnson develops biomanufacturing and pharmaceutical manufacturing methods that connect physical experiments, sensors and computation. His research includes biosensors, bioprocess monitoring and control, theory-guided machine learning and autonomous experimentation. The Advanced Biomanufacturing and Biosensing Lab investigates how measurements can guide experiments and manufacturing decisions, with work spanning functional materials and nonplanar 3D printing.

RESEARCH AND RECOGNITION · 2022–23

His NSF CAREER award supports combining physical models and machine learning to improve biosensor reliability. A subsequent paper in Biosensors and Bioelectronics investigates dynamic sensor signals and theory-guided learning for improved accuracy and speed.

A conversation starter
How can a laboratory use sensor feedback to decide what experiment to run next?

Faculty profile · Lab / personal site · Publications · Highlight source



Zhenyu “James” Kong

Ralph H. Bogle Professor

Intelligent sensing and control for manufacturing

Kong develops AI and machine-learning methods that connect manufacturing processes with computation. His research spans real-time sensing, process monitoring, control and the diagnosis of complex systems. Through the SMART Laboratory and interdisciplinary collaborations, he studies how sensor streams can reveal emerging problems and guide production decisions. Recent projects connect this expertise with semiconductor fabrication and advanced metal manufacturing.

RESEARCH INVESTMENT · 2026

Kong is a co-principal investigator on Virginia Tech’s NSF-supported, $20 million Programmable Cloud Laboratory project for semiconductor research. The university-wide project aims to connect fabrication, characterization and computing through remotely accessible, AI-enabled workflows.

A conversation starter
What must a manufacturing system sense before it can adjust its process intelligently?

Faculty profile · Lab / personal site · Publications · Highlight source



Prahalada Rao

Associate Professor

Sensing and analytics for additive manufacturing

Rao studies manufacturing processes through the combination of sensing, modeling and analytics. A central interest is additive manufacturing: how to monitor a build, detect flaws and control a process while it is running. His work connects physical measurements with statistical and machine-learning methods. This offers students a route from laboratory experiments to algorithms that support more reliable production.

RESEARCH RESULT · OCTOBER 2025

A Rao-led Materials & Design study combines sensing and artificial intelligence to detect flaws during wire-arc additive manufacturing. The university feature explains how monitoring metal as it is printed could support more reliable production of submarine and aircraft components.

A conversation starter
Which sensor signals reveal a developing defect early enough to change the manufacturing process?

Faculty profile · Lab / personal site · Publications · Highlight source



John P. Shewchuk

Associate Department Head · Associate Professor

Manufacturing with reuse and environmental goals

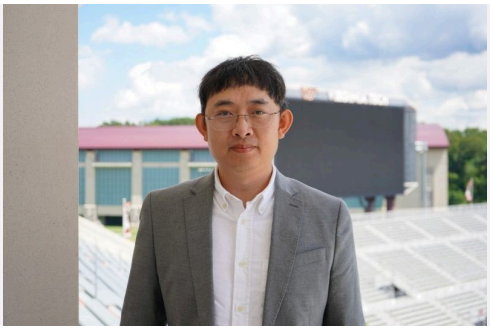
Shewchuk studies sustainable production, lean manufacturing and the integration of manufacturing with remanufacturing. His interests include systems that share resources between new and recovered products, reverse supply chains and circular material flows. This work connects production planning with environmental considerations: changes that improve one operational measure may create effects elsewhere. He also serves as Associate Department Head.

RESEARCH FOCUS

His current faculty profile emphasizes green production systems, modifications to lean practices, hybrid manufacturing/remanufacturing and the return of used products or process waste to production. These topics connect shop-floor decisions with a product's wider material cycle.

A conversation starter
How should production planning change when new and returned products compete for the same resources?

Faculty profile



Tao Sun

Research Associate Professor · CCAM Fellow

Microfabrication and interfaces with biological systems

Sun works at the intersection of manufacturing, microelectromechanical systems, neural engineering and biomedical technology. His research interests connect fabrication methods with devices that sense or interact with the body. His background includes materials engineering and biomedical engineering, and his profile identifies him as a research associate professor and CCAM Fellow. The manufacturing-area page also highlights flexible electronics and connected sensors.

SELECTED PUBLICATION · 2023

His work in Acta Biomaterialia reports a flexible iridium-oxide neural electrode for mouse vagus-nerve stimulation. Other representative publications examine conformable interfaces that measure facial strain and textiles for physiological sensing.

A conversation starter
How can fabrication and material choices improve the interface between an electronic device and living tissue?

Faculty profile · Publications



Yuhao Zhong

Assistant Professor

Explainable AI for safer collaborative manufacturing

Zhong develops data-science methods to improve quality, safety and performance in manufacturing. His research combines explainable AI, computer vision and statistics with the study of collaborative robotic systems. He emphasizes human-centered manufacturing and scientific understanding, connecting a model's prediction with the process knowledge needed to interpret it. Students can investigate how to detect unusual behavior, assess quality and integrate autonomous equipment into shared work environments.

RECOGNITION · 2025

His faculty profile records an Outstanding Paper Award in Manufacturing Processes at the SME North American Manufacturing Research Conference. It also lists a 2023 best student paper award from IISE's Data Analytics and Information Systems Division.

A conversation starter
How can an anomaly-detection system explain its warning well enough to support a manufacturing decision?

Faculty profile · Lab / personal site · Publications

PROGRAM AREA · 03

Management and Systems Engineering

Study how complex systems are designed, managed and improved.

Management and Systems Engineering examines systems made up of people, technologies, information, work processes and decisions. The focus is on the interactions among these elements and with the surrounding environment. Graduate research asks how to design, measure, sustain and improve systems whose behavior cannot be understood by considering each component separately.

The area combines quantitative and qualitative approaches. Topics include system architecture, decision analysis, system dynamics, model-based systems engineering, performance measurement, organizational change and continuous improvement. Faculty apply these methods to healthcare, infrastructure, energy, services and other complex engineered and organizational settings.

The profiles illustrate different scales of inquiry: how two people cooperate in a design task; how a team works across professional boundaries; how a hospital improves a process; or how a policy produces delayed effects across a population. Across these questions, careful problem formulation is as important as the choice of analytical method.

Methods you could use

System dynamics; agent-based modeling; behavioral experiments; interviews and field studies; engineering economics; efficiency analysis; decision modeling.

Questions to explore

A possible research question is whether a performance measure encourages the behavior a system actually needs. Another is how modular design changes the coordination required among teams. Both questions connect technical design with organizational consequences.

Graduate preparation and pathways

Explore the Management and Systems Engineering pathways, the M.S. in Systems Engineering and the Mission Engineering certificate with program faculty. Study location and course format should be discussed for the specific program and advisor.

Faculty connections
Dickerson, Kleiner and Lau are profiled in Human Factors Engineering and Ergonomics. Godfrey appears in teaching and professional practice. The index helps locate these additional connections.

Program overview and affiliated laboratories · Current graduate manual



Alkim Avsar

Assistant Professor

How people cooperate in engineering decisions

Avsar studies collaborative and cooperative systems, focusing on how social factors shape engineering decisions. Her methods combine behavioral experiments, agent-based modeling and surveys with ideas from systems engineering, game theory and behavioral science. The research asks why technically similar teams or system designs can produce different outcomes. Students can investigate how risk attitudes, communication and interactions affect coordination and performance in complex systems.

RECENT PUBLICATION · 2024

With Paul Grogan, Avsar published a Journal of Mechanical Design study identifying design strategies and examining their relationship to performance in paired design tasks. Related work explores differences in risk attitudes during collaborative systems design.

A conversation starter
How do differences in risk attitudes change the way a team designs a shared system?

Faculty profile · Publications



Eric Brubaker

Assistant Professor

Collaborative design across disciplinary boundaries

Brubaker develops design theory and methods for work that crosses cultural, disciplinary and organizational boundaries. His research includes human-AI teaming, organizational learning and collaboration around complex societal problems. Methods range from ethnographic fieldwork to mixed-methods studies and experiments. He works with organizations and communities in settings such as transportation, energy, healthcare and water, connecting the study of design practice with new ways to support it.

RECENT PUBLICATION · 2025

A paper with collaborators at ASME's design-engineering conference uses large-language-model agents to simulate design theory. Other 2025 papers address theory development, qualitative methods and the influence of AI on creativity.

A conversation starter
What enables people with different expertise to develop a shared understanding of a design problem?

Faculty profile · Publications



Navid Ghaffarzadegan

Associate Professor

Dynamics of social systems and public decisions

Ghaffarzadegan uses system dynamics and related modeling approaches to study policy and management problems. His work considers feedback, delays and interactions in healthcare, science policy, higher education and service systems. It helps explain why an intervention can have consequences that differ from its immediate intent. His current research also connects generative AI with agent-based models, opening questions about how to represent human decision-making in simulations.

RECENT PREPRINT · 2026

With Goshi Aoki, he studies AI agents acting as policymakers in a simulated epidemic. In the model, prompts incorporating systems-level knowledge improved decision quality and stability. This is a simulation study reported as a preprint.

A conversation starter
Which feedback loops must a policy model include to avoid misleading conclusions?

Faculty profile · Lab / personal site · Publications · Highlight source



Niyousha Hosseinichimeh

Associate Professor · Grado Faculty Fellow

Simulation and collective learning for health systems

Hosseinichimeh develops modeling methods to improve health and healthcare systems. Her research uses dynamic simulation to help stakeholders understand interactions and evaluate decisions. Applications include mental health, infant mortality, emergency-department use and young people's drinking and driving. Her methodological interests include model calibration and group model building: ways to connect quantitative models with the knowledge of people who work within a system.

RECOGNITION · 2025

The department's honors listing recognizes her with the IISE Society for Engineering and Management Systems teaching award in 2025. Her research provides a connection between systems methods, stakeholder participation and health-policy questions.

A conversation starter
How can a model combine data with stakeholder knowledge while remaining testable and transparent?

Faculty profile · Publications · Highlight source



Taylan G. Topcu

Assistant Professor

Architecture and measurement of complex systems

Topcu combines systems engineering, microeconomics and data science to investigate complex system design. His interests include system architecture, modularity, productivity and the management of safety-critical systems. He uses empirical and mixed-methods research to connect theory with organizational practice. He directs the Systems Engineering master's program and coordinates the Mission Engineering graduate certificate, linking his research with graduate education in system design and assessment.

RECENT PUBLICATION · 2025

A Systems Engineering paper with collaborators examines whether large language models can produce expert-like systems-engineering artifacts and characterizes failure modes. Related work studies digital twins, architecture tradeoffs and reinforcement learning for systems architecting.

A conversation starter
How do we evaluate an AI-generated engineering artifact before relying on it in a complex project?

Faculty profile · Lab / personal site · Publications



Konstantinos P. Triantis

John Lawrence Professor

Performance, efficiency and resilient infrastructure

Triantis studies how engineered and organizational systems perform, particularly when their components are interdependent. His research includes productivity analysis, data envelopment analysis, engineering economics and critical infrastructure. He also studies resilience, sustainability, evacuation planning and healthcare systems. These interests connect operations research with systems thinking: a performance measure should reflect both what a system produces and the conditions under which it operates.

RESEARCH FOCUS

As a co-director of the Systems Performance Laboratory, Triantis connects measurement methods with complex adaptive systems and infrastructure questions. His profile describes work on dynamic performance evaluation and collaborative design.

A conversation starter
How should performance be measured when resilience and efficiency point toward different decisions?

Faculty profile · Lab / personal site · Publications



Eileen M. Van Aken

Professor and Department Head

Organizational performance and sustained improvement

Van Aken studies performance measurement, organizational change and process improvement. Her research examines how organizations define success, implement change and maintain improvements over time. Interests include lean work systems, strategic management and team-based work. As department head, she also connects graduate education with the broader ISE community. Her research offers a route for students interested in understanding how technical improvement methods work within real organizations.

RECENT PUBLICATIONS · 2025

Her work with Kimberly Harry and Wiljeana Glover reviews Kaizen events in hospitals and the factors associated with their success. Other recent work examines lean management in local government and the implementation of performance-measurement systems.

A conversation starter
What makes an improvement effort become part of everyday work rather than a temporary project?

Faculty profile · Publications



G. Don Taylor

Executive Vice Provost · Charles O. Gordon Professor

Logistics and the flow of materials

Taylor's research applies simulation and optimization to logistics, freight transportation and production systems. His work links the movement of materials with the design and management of complex operations. His faculty profile describes extensive collaboration with industry and professional service in industrial engineering. He serves as Executive Vice Provost and Charles O. Gordon Professor, bringing a connection between the department's research traditions and university leadership.

RECOGNITION · 2023

Taylor received IISE's Frank and Lillian Gilbreth Industrial Engineering Award, the institute's highest honor. His research interests connect management and systems engineering with operations research and logistics.

A conversation starter
How can simulation help a logistics organization test a decision before changing its operations?

Faculty profile · Highlight source

PROGRAM AREA · 04

Operations Research

Build mathematical tools for better decisions.

Operations Research uses mathematical models, data and algorithms to support decisions about complex systems and scarce resources. Graduate work can emphasize theoretical methodology, computational algorithms or an application that requires new analytical tools. The area brings together optimization, probability, statistics, simulation and learning.

Virginia Tech's faculty work spans integer and nonlinear programming, stochastic and robust optimization, simulation metamodeling, reinforcement learning, scheduling and healthcare analytics. These methods address decisions in which choices interact, information is incomplete, or outcomes unfold over time. A central research challenge is to understand both what a model recommends and how reliable that recommendation is.

The profiles show a spectrum from mathematical proofs to policy-oriented analysis. Students might investigate the structure of a difficult optimization problem, design a more efficient learning algorithm, or build a model that helps allocate healthcare resources. Cross-area collaborations connect these methods with manufacturing, systems engineering and A2MR.

Methods you could use

Mathematical programming; algorithm design and complexity; stochastic processes; simulation experiments; game theory; statistical learning; computational studies.

Questions to explore

A possible project could compare allocation rules under uncertain demand, or prove when a reformulation makes an optimization problem easier to solve. Application-driven and theory-driven projects can inform one another.

Graduate preparation and pathways

Useful preparation includes mathematical reasoning, probability, linear algebra and programming. Discuss the balance of theory, computation and domain knowledge with potential advisors. The graduate manual provides the requirements for each degree plan.

Faculty connections
Sarin also works in manufacturing. Bansal, Toy, Kannan and Khodadadian are listed with A2MR. Ellis and Taylor provide further connections to applied optimization and logistics.

Program overview and affiliated laboratories · Current graduate manual



Manish Bansal

Associate Professor

Optimization when information is uncertain

Bansal develops mathematical methods for decisions that combine discrete choices, uncertain information and competing interests. His work brings together stochastic and distributionally robust optimization, game theory and machine learning. Applications include supply chains, logistics, network design, aerospace structures and telerobotics. A central question is how to make a useful decision when the probability model itself is uncertain. Students interested in rigorous algorithms can connect mathematical analysis with computational models of real operations.

SELECTED RESEARCH · 2022

With Harsha Gangammanavar, Bansal developed a stochastic decomposition method for two-stage distributionally robust optimization, published in SIAM Journal on Optimization. The work connects uncertainty modeling with algorithms for solving large decision problems.

A conversation starter
How can an optimization model protect a decision against errors in its probability assumptions?

Faculty profile · Publications



Esra Buyuktahtakin Toy

Professor

Learning and optimization for sequential decisions

Toy directs the Systems Optimization and Machine Learning Lab, where researchers combine optimization with learning to support decisions made over time. Her work spans stochastic integer programming, reinforcement learning and game-theoretic models. Applications connect mathematical methods to epidemic response, healthcare logistics, invasive species management and sustainable resource allocation. The common challenge is choosing actions today while accounting for uncertain future conditions and the responses of other decision-makers.

RECENT PUBLICATION · 2026

A Computers & Operations Research paper with collaborators investigates how large language models can help discover heuristics for mixed-integer programming, using single-machine scheduling as the application.

A conversation starter
Where can learning help an optimization algorithm, and where are mathematical guarantees essential?

Faculty profile · Lab / personal site · Publications · Highlight source



Xi Chen

Associate Professor

Learning about complex systems through simulation

Chen studies how to design simulation experiments, extract information from their output and use that information to make better decisions. Her methods combine probability, statistics and optimization. Instead of treating a simulator as a black box that must be run endlessly, her research builds statistical approximations and guides where additional experiments will be most informative. This creates connections among uncertainty quantification, computational efficiency and the comparison of alternative system designs.

RECENT PUBLICATION · 2025

With Jingtao Zhang, Chen published work on multilevel Monte Carlo metamodeling for variance-function estimation in SIAM/ASA Journal on Uncertainty Quantification. The topic extends simulation modeling beyond estimating an average response.

A conversation starter
How should we spend a limited simulation budget to learn about both performance and uncertainty?

Faculty profile · Lab / personal site · Publications



Robert Hildebrand

Associate Professor · Graduate Recruiting Committee Chair

The mathematics behind difficult optimization problems

Hildebrand studies mixed-integer nonlinear optimization: problems in which discrete choices interact with nonlinear relationships. His work examines algorithm design, mathematical formulations and computational complexity. These foundations help explain when a problem can be solved efficiently, when approximation is useful and how a better formulation can improve computation. His interests connect optimization theory with game theory and practical operations-research applications. He also chairs the department's Graduate Recruiting Committee.

RECENT PREPRINT · OCTOBER 2026

A new preprint gives a short proof of a range-relative approximation scheme for polynomial integer optimization in fixed dimension. The result is theoretical; the paper is identified as a preprint, rather than a peer-reviewed publication.

A conversation starter
What makes a formulation computationally difficult, and how can geometry reveal a better algorithm?

Faculty profile · Lab / personal site · Publications · Highlight source



Rohit Kannan

Assistant Professor

Reliable optimization informed by data

Kannan develops methods for global optimization and decision-making under uncertainty, including approaches that use machine learning to improve optimization. His research connects mathematical guarantees with computational performance and applications in energy and process systems. Students can explore how to make decisions using imperfect data, how to handle nonlinear constraints, or how to improve the search for globally optimal solutions. The emphasis is on using learning in ways that remain grounded in optimization theory.

RECOGNITION · 2026

Virginia Tech's research-awards listing identifies Kannan as a 2026 NSF CAREER recipient for work on machine-learning-enhanced branch-and-bound algorithms for scalable global optimization.

A conversation starter
How can learned guidance speed up an optimizer while preserving confidence in its answer?

Faculty profile · Lab / personal site · Publications · Highlight source



Sajad Khodadadian

Assistant Professor

Foundations of reinforcement learning

Khodadadian investigates how learning algorithms make sequential decisions and how reliably they converge. His research brings together reinforcement learning, nonconvex optimization and stochastic approximation. This work is relevant when an agent must learn from experience rather than from a complete model of its environment. A research project may focus on proving performance guarantees, understanding the effect of noise or developing algorithms that use data more effectively.

RECENT PREPRINT · 2025

With Martin Zubeldia, he develops high-probability error bounds for stochastic approximation with Polyak averaging. The preprint gives a general framework and applications to temporal-difference learning and Q-learning.

A conversation starter
How much experience does a learning algorithm need before its decisions become dependable?

Faculty profile · Lab / personal site · Highlight source



Jiaxiang “Jason” Li

Assistant Professor

Optimization for modern machine learning

Li’s research connects continuous optimization with machine learning and deep learning. His recent work includes training large language models, learning from human preferences, bilevel optimization and optimization on curved spaces. These topics link the mathematical design of an algorithm to the computing resources needed to train useful models. His website reports that he joined Virginia Tech in August 2026, following a research-scientist position at Meta.

RECENT PUBLICATIONS · 2026

His website lists an ICML 2026 paper on minimalist optimizer design for language-model pretraining and a TMLR paper on adaptive rank control for robust reinforcement learning. He also received the INFORMS Computing Society Prize in 2024.

A conversation starter
What mathematical structure can make model training more efficient or improve alignment with human feedback?

Faculty profile · Lab / personal site · Publications · Highlight source



Subhash C. Sarin

Paul T. Norton Endowed Professor

Scheduling and the operation of production systems

Sarin develops models and algorithms for production scheduling, sequencing and manufacturing-system control. His research connects applied mathematical programming with the operational decisions that determine how work moves through a facility. Problems include coordinating multiple stages, dividing production lots and scheduling under uncertainty. This work bridges operations research and manufacturing: a mathematical formulation must reflect the actual constraints of machines, materials and production flow.

SELECTED RESEARCH

His publication record includes work on lot streaming in hybrid flow shops, alternative traveling-salesman formulations and risk-sensitive stochastic scheduling. These examples show how foundational optimization methods can support detailed production decisions.

A conversation starter
How can a scheduling model capture the constraints that matter most on a real production floor?

Faculty profile · Publications



Sait Tunc

Assistant Professor

Healthcare decisions and allocation mechanisms

Tunc studies healthcare operations and medical decision-making using stochastic models, queueing, Markov decision processes and game theory. His work examines how system rules and individual behavior interact, particularly when scarce resources must be allocated. The Healthcare Analytics Research Group connects mathematical models with questions about access, screening and organ transplantation. Students can investigate both the design of a policy and how people may respond to its incentives.

RESEARCH RESULT · OCTOBER 2026

A Virginia Tech report describes his team's kidney-matching study in Management Science. Simulations projected more than 900 additional transplants annually under a voluntary targeted-priority mechanism. These are modeled benefits, not observed clinical outcomes. The work was supported by his 2024 NSF CAREER award.

A conversation starter
Can carefully designed incentives improve allocation without requiring everyone to join a new program?

Faculty profile · Lab / personal site · Publications · Highlight source



Huaiyang Zhong

Assistant Professor

Analytics for health and public policy

Zhong combines data analytics, optimization and empirical operations management to inform healthcare and policy decisions. His work connects predictive models with questions about interventions, resource use and the consequences of policy changes. University profiles describe applications in infectious-disease control and the interaction of mental and physical health. This is a route for students interested in quantitative methods whose research questions begin with a population, a health system or a public decision.

NIH-SUPPORTED PROJECT · SEPTEMBER 2026

Zhong leads a project using machine learning and clinical text to improve chronic-pain assessment and identify risks among older rural adults. The research aims to support clinical decisions and more personalized care through collaboration with clinicians and health researchers.

A conversation starter
How can we distinguish a useful prediction from evidence that a policy will change an outcome?

Faculty profile · Publications · Highlight source

PROGRAM AREA · 05

AI, Autonomy, Machine Learning, and Robotics

Connect learning and decision-making with intelligent machines and people.

A2MR brings together research on intelligent systems across the department. It connects learning, optimization, control, perception and robotics with the study of how people interact with technology. Students can pursue foundational methods as well as the design and evaluation of systems intended to operate in complex environments.

The program connects mathematical and computational approaches with physical experimentation and human-system evaluation. Research directions include reinforcement learning, adaptive control, robot perception, collaborative manufacturing, human-AI teams and autonomous experimentation.

This is also a way to connect strengths that appear elsewhere in the booklet. For example, research on process sensing and closed-loop experimentation sits alongside optimization algorithms and the evaluation of human-robot interaction. Students should use the faculty profiles to identify the methods and application settings that best fit their interests.

Methods you could use

Machine learning; optimization; control theory; perception; robotics; human-system experiments; integrated computational and physical testing.

Questions to explore

A possible research question is how an autonomous system should learn and adapt while respecting constraints. Another is how a human teammate understands the system's actions. These questions require attention to the algorithm, its environment and its users.

Graduate preparation and pathways

A2MR is an ISE Ph.D. concentration at Blacksburg, with related M.S. and M.Eng. focus areas. Students develop an individualized plan with their advisor and advisory committee; the current program page describes the concentration and its research connections.

Faculty connections
The next page profiles Andrea L'Afflitto and points to the other faculty listed on the A2MR page, whose full profiles appear earlier in this booklet.

Program overview and affiliated laboratories · Current graduate manual



Andrea L'Afflitto

Associate Professor

Control and autonomy for robotic systems

L'Afflitto studies autonomous robots, unmanned systems and control theory. His research connects mathematical analysis with the design of systems that must operate robustly in uncertain environments. The A2MR program identifies adaptive control, data-driven methods and robotics among his interests. His experience spans aerospace systems and aerial robotics, offering students a route between theoretical control methods and physical autonomous systems.

RECOGNITION · 2025

The department's honors page lists L'Afflitto as a 2025 AIAA Associate Fellow. His faculty profile also records a 2018 DARPA Young Faculty Award. He is a faculty connection for both manufacturing and the A2MR concentration.

A conversation starter
How can a robot adapt to uncertainty while respecting the constraints required for safe operation?

Faculty profile · Lab / personal site · Highlight source

A2MR CONNECTIONS

A shared faculty network

The A2MR program lists the following additional faculty. Their full profiles appear in other sections.

Manish Bansal	21
Eric Brubaker	16
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Myounghoon "Philart" Jeon	05
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A2MR faculty and laboratories

Faculty contributing to teaching, applied projects and professional education.



Kathleen Carper

Collegiate Assistant Professor

Technical communication in the ISE community

Carper is a collegiate assistant professor in Technical Communications. Her teaching connects engineering analysis with the writing needed to explain problems, present evidence and communicate decisions. Clear technical communication helps students make their engineering work understandable to collaborators and other audiences.

TEACHING CONNECTION

Carper teaches in the department's Technical Communications area, supporting the writing and communication skills that accompany engineering analysis, design and professional practice.

A conversation starter
What communication habits make technical reasoning easier for collaborators and decision-makers to follow?

Faculty profile



Natalie Cherbaka

Undergraduate Program Director · Collegiate Professor

Production systems and experiential engineering education

Cherbaka is the Undergraduate Program Director and a collegiate professor. Her teaching spans production planning, facilities and material handling, engineering economics and systems design. Her experience includes manufacturing consulting and production planning at IBM. The manufacturing-area page lists interests in warehousing, supply chains and logistics applications. She connects the department's applied systems expertise with the design of learning experiences.

RECOGNITION · 2025

The department lists Cherbaka as a Diggs Teaching Award recipient in its 2025 faculty recognition. Ellis's faculty profile also lists a 2026 in-press education paper with Cherbaka on an online production-planning game.

A conversation starter
How can a simulation or game help students understand the consequences of production decisions?

Faculty profile · Highlight source

Faculty contributing to teaching, applied projects and professional education.



Kelsey Coleman

Associate Professor of Practice · Director, Learning Factory

Industry partnerships and the Learning Factory

Coleman is an associate professor of practice and director of the Learning Factory. Her work connects the department with industry through consulting projects, senior design partnerships and workforce development. Her professional background includes operations management, continuous improvement, supply-chain analysis and manufacturing engineering at Eaton. These experiences connect industrial-engineering methods with the practical constraints faced by organizations.

PROFESSIONAL PRACTICE

Her faculty profile describes leadership of the Learning Factory and Center for High Performance Manufacturing, including external-project coordination, continuing professional education and the center's daily operations.

A conversation starter
What makes an industry problem a useful project for applying and testing industrial-engineering methods?

Faculty profile



Joseph Godfrey

Collegiate Assistant Professor · Co-director, Systems Performance Laboratory

Systems theory and emerging technologies

Godfrey is a collegiate assistant professor and co-director of the Systems Performance Laboratory. His interests include social choice, cooperative games, systems theory, engineering economics, data science and quantum systems engineering. His teaching responsibilities span decision analysis, technology transfer, project management and AI for systems. His profile describes work on quantum workforce development and the integration of AI into organizational processes.

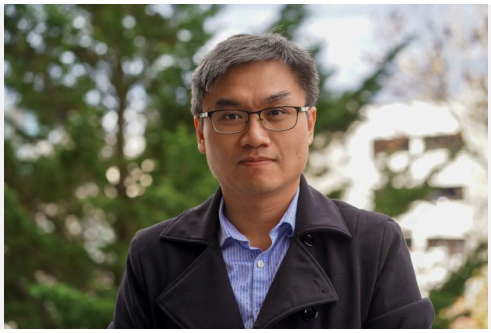
PROFESSIONAL AND TEACHING FOCUS

Godfrey connects systems concepts with the development and use of emerging technologies. His public profile also describes activity around a Virginia quantum network and collaborations on education for the quantum workforce.

A conversation starter
How should an organization evaluate a new technology when its technical and workforce requirements are evolving?

Faculty profile · Lab / personal site

Faculty contributing to teaching, applied projects and professional education.



Win PV Nguyen

Collegiate Assistant Professor

Practical engineering tools and responsible AI use

Nguyen is a collegiate assistant professor whose interests include generative AI in learning, task-specific AI agents and the rapid development of industrial-engineering tools. His work connects programming, spreadsheets and simulation with practical process and decision analysis. He also studies data related to student success, retention and timely graduation. His teaching spans facilities, logistics, production planning and quality control.

CURRENT INTERESTS

His faculty page highlights responsible and effective AI use, prompt design and customized workflows. It also describes experience in software automation and manufacturing-process analysis, linking classroom tools to engineering practice.

A conversation starter
How can a customized AI or software tool improve a workflow while keeping its reasoning checkable?

Faculty profile



Laura Savage

Collegiate Associate Professor

Engineering economics and manufacturing education

Savage is a collegiate associate professor in the department's manufacturing education community. Her teaching connects engineering fundamentals with the economic evaluation of decisions. She coauthored an Engineering Economy digital textbook with Chris Kwaramba, listed among the department's spring 2026 course materials. Her contributions include classroom instruction and the development of resources that help students apply industrial-engineering concepts.

TEACHING RECOGNITION · 2024

Savage received a College of Engineering Certificate of Teaching Excellence in the department's spring 2024 honors listing. Her promotion to collegiate associate professor was announced in 2025.

A conversation starter
How does graduate preparation connect manufacturing fundamentals with the analysis of integrated production systems?

Faculty profile · Highlight source

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Keep exploring

Department directory · Graduate application guidance

Profile links provide the source for research summaries and selected publications. Additional highlight sources are linked within profiles. Suggested conversation questions are editorial prompts.

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HFEE Human Factors Engineering and Ergonomics · **MFG** Manufacturing Systems Engineering
MSE Management Systems Engineering · **OR** Operations Research
A2MR AI, Autonomy, Machine Learning, and Robotics
T/P Teaching and professional practice (includes teaching interests)

Accessibility and assistive technology

Myounghoon "Philart" Jeon HFEE · 05
Sol Lim HFEE · 08

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Autonomy, robotics and control

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Sound, hearing and sensory perception

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